GG-V-350
AUGUST 9, 1960
SUPERSEDING
Fed. Spec. GG-I-450
November 19, 1947

FEDERAL SPECIFICATION

VIEWERS, DENTAL RADIOGRAPHIC FILM, 110 VOLT, AC-DC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers dental radiographic film viewers, suitable for viewing dental radiographs with or without mounts.

1.2 Classification.

1.2.1 *Types.*—Film viewers shall be of the following types, as specified (see 6.1):

Type I. — Magnifying, for viewing films up to and including 5- by 7-inch film.

Type II. — Adjustable viewing surface viewer for 8- by 10-inch film in vertical or horizontal plane.

Type III.—Nonadjustable viewing surface viewer for 8- by 10-inch film in vertical or horizontal plane.

Type IV.—Adjustable angle viewer with 5½- by 14-inch viewing surface.

2. APPLICABLE SPECIFICATIONS STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following speecifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

QQ-S-766 — Steel Plates, Sheets, and Strip—Corrosion Resisting.

QQ-S-781—Strapping, Flat; Steel.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636—Box, Fiberboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipments (Civilian Agencies).

Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547 — Liners, Case, Waterproof.

570397/6076

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Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** — The following publications form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

National Electrical Code.

(The National Electrical Code is published by the National Board of Fire Underwriters, 85 John Street, New York 7, New York, 222 West Adams Street, Chicago 6, Ill., and 465 California Street, San Francisco 4, Calif.)

Underwriters' Laboratories, Inc. Standards.

(Underwriters' Laboratories Inc. standards are available from the Underwriters' Laboratories, Inc., 161 Sixth Ave., New York 13, N. Y., 207 East Ohio Street, Chicago 11, Ill., and 1655 Scott Blvd., Santa Clara, Calif.)

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 Each bidder shall submit to the contracting agency proof that the equipment he proposes to supply under this specification conforms to the requirements of Underwriters' Laboratories, Inc. The label or listing of the Underwriters' Laboratories, Inc. may be accepted as evidence that the equipment conforms to this requirement.

3.1.2 In lieu of the label or listing, the bidder may submit independent proof satisfactory to the contracting agency that his equipment conforms to the published standards, including methods of tests, of the Underwriters' Laboratories, Inc.

3.1.3 Compliance with Underwriters' Laboratories, Inc. requirements regarding fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his commodity.

3.2 Material.—Materials used in the fabrication of the viewers shall be of the highest quality throughout and suitable in all respects for the purpose intended. All wiring and switches shall conform to the requirements of the National Electrical Code and Underwriters' Laboratories, Inc. standards.

3.2.1 *Housing.*—The body, ends, support and lens housing, when applicable, shall be sheet steel, brass, or aluminum.

3.2.2 *Slide rods (type I only).*—The slide rods shall be corrosion-resisting steel, class 430 of Federal Specification QQ–S–766.

3.2.3 *Lens bezel and binding (type I only).* —The lens bezel and binding shall be brass or other suitable metal, chromium-plated.

3.3 Type I viewer.

3.3.1 *Design.* — The viewer shall be designed for viewing dental radiographic films, with or without mounts, as well as general purpose X-ray and photoroentgenographic films, up to and including 5- by 7-inch size. The viewer shall be suitable for operation on 110 volts, 40 watts, either alternating or direct current. A tilting arrangement shall be provided to permit positioning of the viewing surfaces and magnifying glass at any desired position in a 180-degree arc. The viewer shall consist essentially of the viewing box and the magnifying glass, and shall be equipped with an incandescent lamp, rheostat, "on-off" switch, and cord and attachment plug cap.

3.3.2 *Viewing box.* — The cylindrical viewer housing shall be approximately 14¾ inches long and 6¼ inches in diameter. An incandescent lamp shall be mounted in the back. There shall be a groove approximately in the center for holding the plastic viewing panel which shall be approximately 14½ by

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5 3/16 inches by 1/8 inch. Directly in front of the viewing panel shall be a similar groove for inserting film mounts or films. The left end of the cylindrical housing shall be provided with an opening through which film mounts or films can be inserted. In the front half of the housing shall be two nonreflecting slide rods which shall accommodate the carriage for the magnifying glass and its mounting tube. This carriage shall move the entire length of the viewer housing. The electrical connections, toggle switch, rheostat control, and extension cord shall be located on the left end of the cylindrical housing. On the opposite end, provision shall be made for a pivoting arrangement between the stand and the case; a locking device shall be provided to securely lock the stand in any desired position. The viewing window shall be a sheet of white translucent plastic 0.125 ± 0.015 inch thick with a light transmission factor of approximately 55 percent.

3.3.3 *Magnifying glass.*—The magnifying glass shall consist of a lens attached to a holding tube by means of a bezel and binding. The magnifying lens shall be not less than 3½ inches in diameter. The tube holding the lens shall be approximately 4 inches long and so constructed that it can be removed from the carriage, allowing an unobstructed view of a complete 14-film mount without magnification. It shall also be adjustable in the carriage to allow for various degrees of magnification. The focal length of the lens shall be approximately 8 inches to produce minimum magnification of 1½.

3.3.4 *Incandescent lamp.*—Lamp shall be tubular type, incandescent, 115 volts, 40 watts, bulb T–8 and coated. The lamp shall have a disk base and C–8 type filament and shall have an approximate 1500-hour life. The overall length of the lamp shall be 11¾ inches (see 6.2).

3.3.5 *Rheostat.*—The rheostat shall be of the carbon pile or wire wound type, attached to the left side of the viewing box, and shall be capable of varying the intensity of light from "off" to "full on". The rheostat shall be of appropriate resistance and power ratings to vary the light intensity.

3.3.6 *Cord and attachment plug cap.*—Each viewer shall be equipped with a suitable, flexible, rubber-covered cord of sufficient length to extend beyond the viewer a minimum distance of 6 feet. The cord shall be equipped with a two-pole, parallel-blade, nonpolarized attachment plug cap rated 10 amperes at 250 volts.

3.4 **Type II and type III viewers.**

3.4.1 *Design.*—Type II and type III viewers shall be designed for viewing single 8- by 10-inch films and dental films in dental mounts up to 18 in number. When mounted on a wall, or standing on a desk in a horizontal or vertical position, an entire 8- by 11-inch viewing area shall be available for use. The type II viewer shall be furnished with adjustable masking shutters to permit viewing areas from 2 inches square up to and including 8- by 10-inch film size.

3.4.2 *Main body.*—The main body of the viewer shall contain all the electrical components necessary for the operation of the viewer. The overall dimensions of the body shall not exceed 16¾ inches wide by 12½ inches high by 5½ inches deep. The body shall have a viewing panel made of white translucent plastic and shall be 12 5/16 inches wide by 9⅜ inches high by 1/8 inch thick. The light transmission factor shall be approximately 55 percent. The plastic panel shall be free from any defects, such as bubbles, variations in thickness, and flaws in finish. The viewing area shall be provided with film holders on two adjacent sides to hold films when the viewer is used in horizontal or vertical position. For the type II viewer, the viewing area shall be furnished with concealed masking shutters for varying viewing areas in two directions, starting with a 2-inch square in one corner when fully closed, to an area of 8 by 11 inches when fully opened. The shutters shall be controlled by knobs, one for each shutter, and so located that the viewer may

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be used in horizontal or vertical position without resting on the knobs.

3.4.3 *Electrical components.*—The viewers shall be furnished with one 22-watt fluorescent lamp. Ballast shall be of the instant start type, designed to operate one 22-watt fluorescent lamp. The lamp shall be located behind the viewing panel, and shall be easily removable from its socket for replacement. A suitable "ON"-"OFF" switch and the necessary lamp auxiliaries for operation on 115 volt, alternating current, shall be provided. The switch may be mounted on the front, top, or side of the viewer in such a manner as to prevent water or solutions from coming in contact with it. The wiring shall consist of a 2-wire cord set extending at least 6 feet from the viewer.

3.4.4 *Illumination.*—With the lamp operated at a rated voltage, the brightness at the center of viewing area shall not be less than 400-foot lamberts. The total variation in brightness along any horizontal or vertical line on the surface of viewing area shall be not greater than 25 percent of the maximum brightness.

3.5 Type IV viewer.

3.5.1 *Design.*—Type IV viewer shall be designed to provide illumination of sufficient intensity, proper color, and uniform distribution to permit diagnosis of dental radiographs. The viewer shall consist of a housing, viewing panel, trim flanges, removable mask, adjustable base, and necessary lighting, wiring, and switch.

3.5.2 *Housing.*—The viewer housing shall be of metal construction 15¼ inches wide by 6½ inches high by 7 inches deep. It shall be a modified triangular section with an opening for the viewing panel, and so designed that the viewing panel shall be at a 45-degree angle with respect to the horizontal.

3.5.3 *Base.*—The viewer shall have a metal adjustable base which shall permit angulation of the viewer housing so that the viewing panel may be maintained at any desired angle with respect to the horizontal between 75 degrees and 15 degrees, plus or minus 5 degrees. The base shall be rectangular in shape and be provided with four rubber feet to prevent marring of surfaces.

3.5.4 *Trim flanges and mask.*—The viewer shall be provided with three chromed-steel trim flanges, positioned around the viewing area to fit over the side and bottom edges of the plastic viewing panel. The bottom flange shall be so designed to provide a ledge on which to rest the dental mounts. A chromed-steel mask 15 inches long shall be provided which can be slipped into the upper end of the two side trim flanges to restrict the illuminated field to 13$^{5}/_{16}$ inches wide by 3$^{15}/_{16}$ inches high.

3.5.5 *Viewing panel.*—The viewer shall be provided with a durable, white, translucent plastic panel to provide well-diffused lighting, devoid of high spots. The panel shall be 14 inches long by 5½ inches wide by ⅛ inch thick. The panel shall be positioned over the viewing section of the housing and maintained in place by trim flanges. The light transmission factor of the panel shall be approximately 55 percent.

3.5.6 *Electrical components.*—One 12-inch, 14-watt fluorescent lamp shall be installed in the housing to illuminate the viewing area at the proper intensity. The lamp shall be energized by a push-button type of "ON-OFF" switch. The wiring shall consist of a two-wire cord set extending at least 6 feet from the viewer. All necessary lamp auxiliary equipment for operation on 115 volts, alternating current, shall be provided.

3.6 Dielectric strength.—There shall be no insulation breakdown in the viewers when tested as specified in 4.4.2.

3.7 Finish.—All metal parts, with the exception of corrosion-resistant steel, or plated steel, shall be finished with acid-, alkali-, and solvent-resisting enamel. All exposed corrodible metal parts shall be chromium-plated and shall show no evidence of corrosion when

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tested as specified in 4.4.1. Crackle type finishes or other rough surfaces difficult to keep clean will not be acceptable.

3.7.1 *Degreasing.* — Metal parts shall be thoroughly cleaned and chemically degreased prior to adding the primer coat.

3.7.2 *Primer.*—After degreasing, a suitable primer coat shall be applied. The primer coat shall form a grounding bond and base for the finished enamel.

3.7.3 *Color.* — Unless otherwise specified, the exterior of the viewers, shall be finished with a baked enamel, in cream white conforming to color 13531 of Federal Standard No. 595 (see 6.1 and 6.3). The interior and reflecting surfaces shall have a white matte baked-enamel finish. Chrome-plated metallic or nonmetallic parts shall not be painted.

3.8 **Identification marking.** — A suitable nameplate shall be permanently attached to each viewer and shall be embossed or indented with the name or registered trademark of the manufacturer and the electrical rating.

3.8.1 *For Military Medical Supply Agency procurement.*—In addition to the marking required in 3.8, the type I viewer shall be permanently marked with the applicable Federal stock number.

3.9 **Service data.**—The contractor shall enclose with each unit two copies each of one or more booklets which contain complete instructions for installation, operation, maintenance, and a list of component parts. Wiring diagrams shall be included for electrical items or components. Each part of the unit, other than common fasteners, etc., shall be identified in the service data and parts list by the manufacturer's part number.

3.9.1 *For Military Medical Supply Agency procurement.*—The contractor shall mail 12 additional copies of the service data required in 3.9 to the contracting officer for distribution.

3.10 **Workmanship.**—Workmanship shall be first class throughout. Viewers shall be free from defects which may impair their serviceability or which detract from their appearance.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Supplier responsibilities for inspection.** —The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Inspection.* — Inspection as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality.*—When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 **Sampling.**

4.2.1 *For examination.*—Sampling for examination shall be conducted in accordance with Military Standard MIL–STD–105.

4.2.2 *For tests.*—Sampling for tests shall be conducted in accordance with Military Standard MIL–STD–105. The acceptance number shall be zero (0) for all sample sizes.

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Test characteristic	Requirement paragraph	Test paragraph	Inspection level
Corrosion	3.7	4.4.1	See 4.2.4
Dielectric strength	3.6	4.4.2	L4

4.2.3 *Raw materials.*—Four ounces of each lot of each metal used in the construction of the viewers shall be selected for chemical analysis.

4.2.4 *Components.*—Sufficient samples of the components shall be selected from each lot of material used in the manufacture of the viewers to perform tests as required by the applicable subsidiary specifications or by tests specified herein.

4.2.5 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 **Examination.** — The viewers shall be examined to determine compliance with all requirements contained in this specification and with the classification of defects, as follows:

4.3.1 *Classification of defects.*

Categories and defects

Major: AQL 1.0 percent defective.

101. Viewers not furnished complete with all specified component parts.
102. Dental films (with or without mounts) and X-ray films up to and including 5- by 7-inch size fail to fit into viewer grooves (type I viewer).
103. Tilting arrangements fail to function as specified in 3.3.2 and 3.5.3.
104. Magnifying glass carriage fails to move smoothly over entire length of viewer housing (type I viewer).
105. Stand fails to lock and unlock properly in any desired position (type I and type IV viewers).
106. Component parts not free of cracks, tears, ruptures, or ripples.
107. Edges not free of burs and sharpness.
108. Component parts of viewer not securely fitted and properly assembled.

Minor: AQL 2.5 percent defective.

201. Dimensions not as specified.
202. Cord length shorter than 6 feet.
203. Nameplate not present, or specified information thereon not legible or complete.
204. Plating and finish not free of blisters and unplated or uncoated areas.
205. Labels or printing on packages not legible and free of defacing marks, scratches, and tears.

4.4 Tests.—Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 *Corrosion test.*—The quality of the chromium plating shall be determined by boiling the parts in distilled water in a glass beaker for 30 minutes and allowing the parts to remain submerged in the water for 24 hours.

4.4.2 *Dielectric strength test.*—A potential of 1,000 volts, 60 cycle, alternating current, shall be applied for 1 minute between the case and each end of the AC plug with the "on-off" switch in the "on" position and the rheostat, if applicable, set for maximum resistance.

5. PREPARATION FOR DELIVERY

For civil agency procurement the levels of preservation, packaging and packing protection used are defined in Federal Standard No. 102.

5.1 **Packaging.**

5.1.1 *Level A or B.*

5.1.1.1 *Type I.*—Each unit shall be disassembled as far as is practicable and packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–636. Carton design shall include interior fittings or sufficient cushioning material to insure against shifting, breakage, or marring of surfaces. Closure shall be adequate to prevent spillage of contents under normal handling.

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5.1.1.2 *Type II and Type III.*—Each unit shall be disassembled and packaged as follows:

(*a*) *Plastic viewing panels:* The viewing panel sheet shall be interleaved with sheets of double-faced corrugated fiberboard of not less than 200-pound Mullen test. The bundle so formed shall be backed on both sides with similar sheets of fiberboard and then taped with either gummed or pressure sensitive tape and either packaged in a fiberboard carton of appropriate size or completely wrapped in flexible corrugated fiberboard. Carton or wrapping shall be secured to prevent accidental opening.

(*b*) *Fluorescent lamp:* The lamp, removed from the illuminator, shall be packaged in a suitable sleeve of corrugated fiberboard and then packed in a fiberboard carton or completely overwrapped in flexible corrugated fiberboard. Carton or wrapping shall be secured to prevent accidental opening.

(*c*) *Illuminator body:* The main body of the illuminator shall have all finished surfaces protected by pads or suitable wrapping.

5.1.1.3 *Type IV.*—Each unit, disassembled insofar as practical, shall be packaged in a fiberboard carton designed with suitable interior fittings to hold all parts securely and safely to prevent damage or breakage. If required, adequate cushioning material shall be placed within the container to prevent movement of the parts. Cartons shall be adequately secured to prevent accidental opening.

5.1.2 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Viewers packaged as specified in 5.1, shall be packed in quantities as specified in table I, in an exterior container constructed in accordance with Federal Specification PPP–B–621, table IV.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.*—Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*—Viewers packaged as specified in 5.1, shall be packed in quantities as specified in table I, in snug-fitting shipping containers conforming to Federal Specification PPP–B–636, Type I or II, class 1, style optional. Sealing and closure of the boxes shall be in accordance with the appendix to the box specification. Strapping when specified, shall be flat steel, conforming to Federal Specification QQ–S–781, Type I, class B.

TABLE I.

Viewers Type	Units per shipping container
I	6
II	1
III	1
IV	4

5.2.3 *Level C.*—The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Civil agencies.*—In addition to any special marking required in the contract or

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purchase order, unit and shipping containers shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military activities.* — In addition to any special marking required in the contract or purchase order, unit and shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 **Ordering data.**—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type required (see 1.2).

(*c*) Color, if other than specified (see 3.7.3).

(*d*) Selection of applicable levels of packaging, packing and marking (see 5.1, 5.2, and 5.3).

6.2 The lamp referenced in 3.3.4 shall be General Electric Co.'s No. L40/W lumiline, or equal.

6.3 **Availability of color chip.**—An individual 3- by 5-inch chip of the color (cream white, No. 13531), referenced in 3.7.3, is available from General Services Administration, Business Service Center, Washington 25, D. C. Price 5 cents.

6.4 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.5 **Transportation description.**—Transportation description applicable to this item is:

Lamps, Electric, Blueprinting or Photographic, NOIBN with or without metal reflectors or stands.

Carload minimum weight 15,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor volume minimum weight 15,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

CUSTODIANS:
Army—Army Medical Service
Navy—Bureau of Medicine and Surgery
Air Force

Copies of this specification may be purchased for 5 cents each.

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